Лабораторная работа 2

Измерение и тестирование пропускной способности сети. Интерактивный эксперимент

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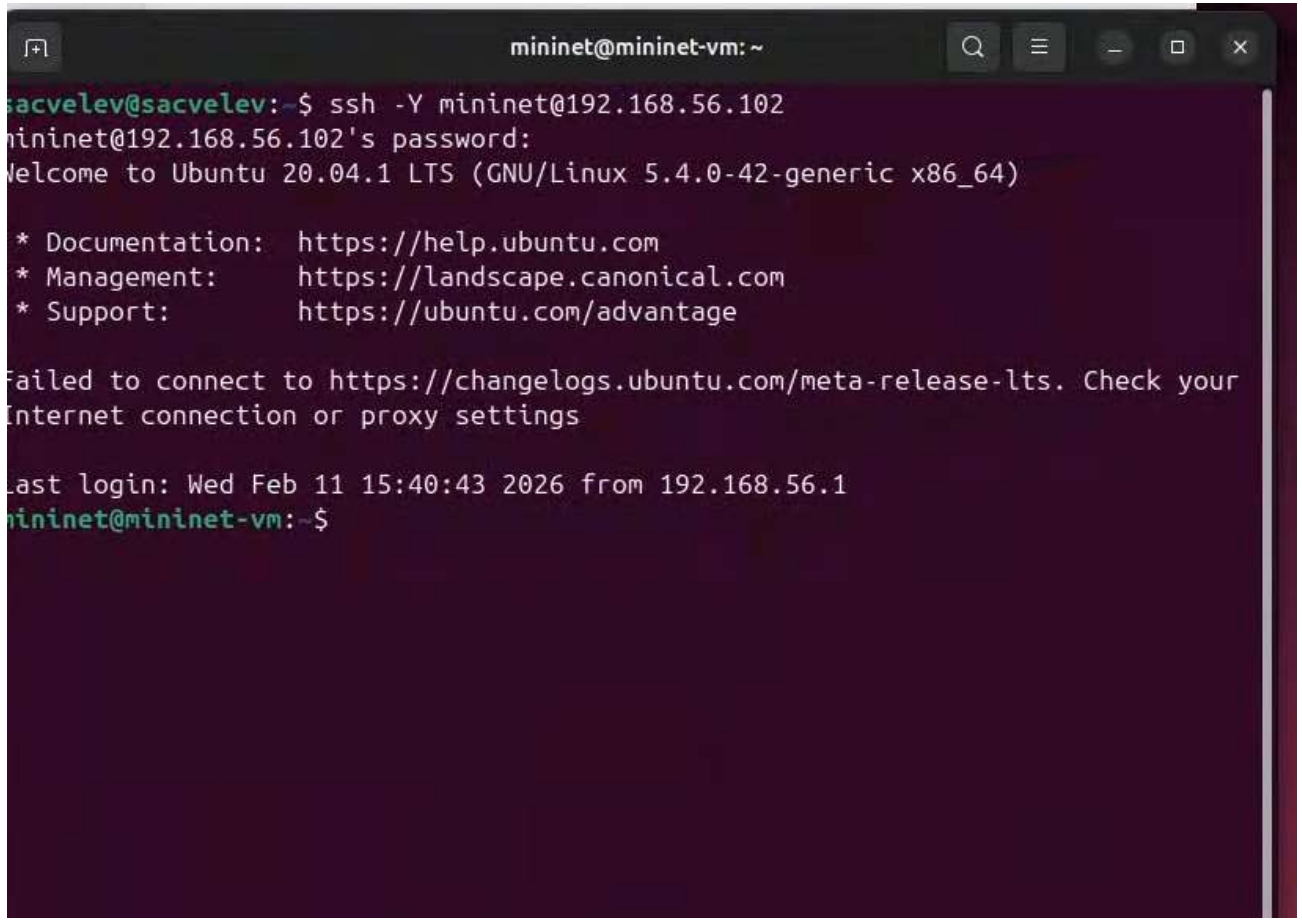
Цель работы

Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

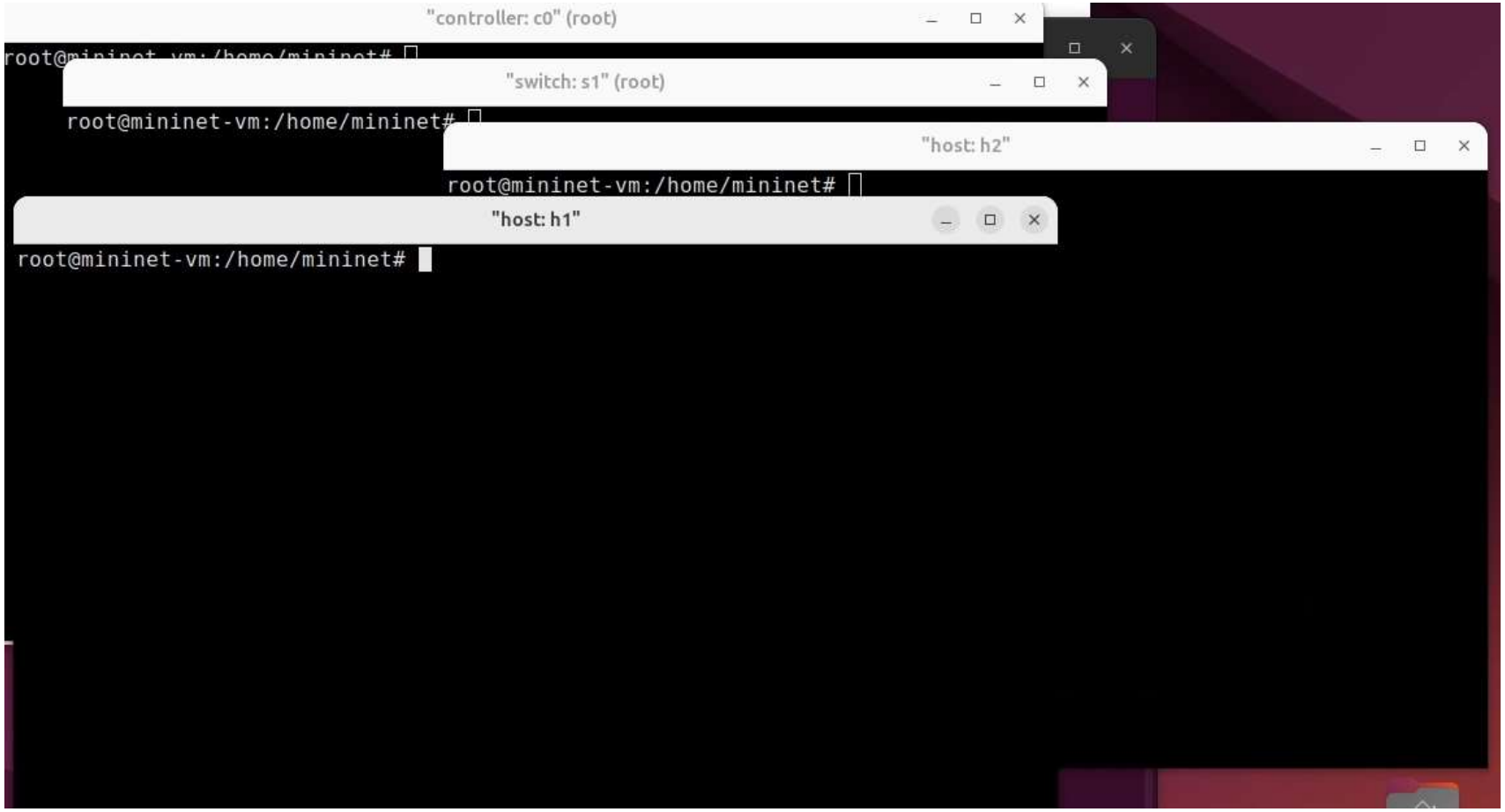
Запуск mininet с хостовой машины



Установка и обновление ПО

```
mininet@mininet-vm:~$ sudo apt-get update
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,56
4 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:5 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [1,11
4 kB]
Get:7 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,9
57 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [881 k
B]
Get:9 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518
kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadat
a [14.4 kB]
Get:11 http://security.ubuntu.com/ubuntu focal-security/restricted i386 Packages
 [42.5 kB]
Get:12 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Package
s [3,768 kB]
49% [7 Packages 1,297 kB/3,957 kB 33%] [12 Packages 3,001 kB/3,768 kB 80%]
```

Создание минимальной топологии



Отправка пакетов данных с h1 к h2

```
mininet@mininet-vm: ~
s1 ...
*** Starting CLI:
mininet> h2 iperf3 -s &
mininet> h1 iperf3 -c h2
Connecting to host 10.0.0.2, port 5201
[  5] local 10.0.0.1 port 56408 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  4.91 GBytes  42.1 Gbits/sec    2   3.58 MBytes
[  5]   1.00-2.00   sec  4.42 GBytes  37.9 Gbits/sec    0   3.58 MBytes
[  5]   2.00-3.00   sec  3.87 GBytes  33.1 Gbits/sec   10   2.52 MBytes
[  5]   3.00-4.00   sec  3.81 GBytes  32.9 Gbits/sec    1   2.54 MBytes
[  5]   4.00-5.00   sec  3.76 GBytes  32.3 Gbits/sec    1   2.55 MBytes
[  5]   5.00-6.00   sec  3.91 GBytes  33.6 Gbits/sec   47   1.88 MBytes
^C[  5]   6.00-6.26   sec  1.01 GBytes  34.1 Gbits/sec    0   1.94 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-6.26   sec  25.7 GBytes  35.3 Gbits/sec   61             sender
[  5]   0.00-6.26   sec  0.00 Bytes  0.00 bits/sec                  receiver
iperf3: interrupt - the client has terminated
mininet> h2 killall iperf3
warning: this system does not seem to support IPv6 - trying IPv4
iperf3: error - unable to start listener for connections: Address already in use
iperf3: exiting
mininet> h2 killall iperf3
iperf3: no process found
mininet>
```

Проверка разных параметров

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -t 5
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 56412 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  4.77 GBytes  41.0 Gbits/sec   15   17.7 MBytes
[  7]   1.00-2.00   sec  4.73 GBytes  40.6 Gbits/sec    0   17.7 MBytes
[  7]   2.00-3.00   sec  4.76 GBytes  40.9 Gbits/sec   93   12.4 MBytes
[  7]   3.00-4.00   sec  4.70 GBytes  40.4 Gbits/sec    0   12.4 MBytes
[  7]   4.00-5.00   sec  4.67 GBytes  40.1 Gbits/sec    0   12.4 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-5.00   sec  23.6 GBytes  40.6 Gbits/sec  108             sender
[  7]   0.00-5.00   sec  23.6 GBytes  40.6 Gbits/sec                  receiver
```

Проверка разных параметров

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -n 16G
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 56420 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  4.67 GBytes  40.1 Gbits/sec   47   2.16 MBytes
[  7]   1.00-2.00   sec  4.74 GBytes  40.7 Gbits/sec   92   1.61 MBytes
[  7]   2.00-3.00   sec  4.73 GBytes  40.6 Gbits/sec    1   1.70 MBytes
[  7]   3.00-3.41   sec  1.86 GBytes  39.5 Gbits/sec    1   1.72 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-3.41   sec  16.0 GBytes  40.4 Gbits/sec  141             sender
[  7]   0.00-3.41   sec  16.0 GBytes  40.3 Gbits/sec                  receiver
```

Проверка разных параметров

```
iperf Done.
root@mininet-vm:/home/mininet# ^C
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -u
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 49973 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  7]   0.00-1.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   1.00-2.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   2.00-3.00   sec   126 KBytes  1.03 Mbits/sec  89
[  7]   3.00-4.00   sec   130 KBytes  1.07 Mbits/sec  92
[  7]   4.00-5.00   sec   127 KBytes  1.04 Mbits/sec  90
[  7]   5.00-6.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   6.00-7.00   sec   127 KBytes  1.04 Mbits/sec  90
[  7]   7.00-8.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   8.00-9.00   sec   127 KBytes  1.04 Mbits/sec  90
```

Проверка разных параметров

```
iperf Done.
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -p 3250
Connecting to host 10.0.0.2, port 3250
[  7] local 10.0.0.1 port 36378 connected to 10.0.0.2 port 3250
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  4.33 GBytes  37.2 Gbits/sec   84   5.53 MBytes
[  7]   1.00-2.00   sec  4.11 GBytes  35.3 Gbits/sec    2   5.53 MBytes
[  7]   2.00-3.00   sec  3.92 GBytes  33.7 Gbits/sec   94   2.80 MBytes
[  7]   3.00-4.00   sec  4.06 GBytes  34.9 Gbits/sec    1   2.87 MBytes
[  7]   4.00-5.00   sec  4.28 GBytes  36.8 Gbits/sec   47   2.09 MBytes
[  7]   5.00-6.00   sec  4.05 GBytes  34.8 Gbits/sec  148   1.10 MBytes
[  7]   6.00-7.00   sec  3.64 GBytes  31.2 Gbits/sec    0   1.20 MBytes
[  7]   7.00-8.00   sec  3.52 GBytes  30.2 Gbits/sec    0   1.28 MBytes
[  7]   8.00-9.00   sec  3.80 GBytes  32.7 Gbits/sec  250    670 KBytes
[  7]   9.00-10.00  sec  3.84 GBytes  32.9 Gbits/sec   63   1.43 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  39.6 GBytes  34.0 Gbits/sec  689             sender
[  7]   0.00-10.00  sec  39.6 GBytes  34.0 Gbits/sec                  receiver
```

Сохранение в json

```
                        "sender":       true
                },
                "sum_received": {
                        "start":        0,
                        "end":  10.0012,
                        "seconds":      10.0012,
                        "bytes":        46733341720,
                        "bits_per_second":      37382187513.498375,
                        "sender":       true
                },
                "cpu_utilization_percent":      {
                        "host_total":   90.994161294137015,
                        "host_user":    1.0445658993709095,
                        "host_system":  89.949605390325075,
                        "remote_total": 35.639243279673465,
                        "remote_user":  1.2465482599549851,
                        "remote_system":        34.392695019718481
                },
                "sender_tcp_congestion":        "cubic",
                "receiver_tcp_congestion":      "cubic"
        }
}
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -J > /home/mininet/work/lab_i
perf3/iperf_results.json
```